

Assignment 5A

Airline Delays

Approach

A small table is portrayed that gives on-time and delayed arrivals for Alaska and AM West at five West Coast airports. The problem is which airline has better reliability and does it depend on aggregation of counts.

In R, I am going to use packages tidyr and dplyr to delete the empty row, fill the missing names of airlines and pivot the data into one line per airline and city. After that, I am going to compare the percentage of delays overall and for each city using summary table and grouped bar chart, validating the numbers against the original source. The difficult parts are the messy wide format of the table that leads to generation of NA values that have to be dealt with and comma-separated numbers that have to be entered manually as integers.

Code Base

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.6.1
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ forcats   1.0.1   ✓ readr     2.2.0
✓ ggplot2   4.0.3   ✓ stringr   1.6.0
```

```
✓ lubridate 1.9.5    ✓ tibble    3.3.1
✓ purrr      1.2.2    ✓ tidyr     1.3.2
```

```
— Conflicts ————— tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
csv_url <- "https://raw.githubusercontent.com/daanishrasheed/DATA607/refs/
heads/main/Assignment%205A/airline_delays.csv"
```

```
raw <- read_csv(csv_url, show_col_types = FALSE)
```

New names:

- `` -> `...1`
- `` -> `...2`

```
raw
```

```
# A tibble: 5 × 7
  ...1    ...2 `Los Angeles` Phoenix `San Diego` `San Francisco` Seattle
  <chr>  <chr>      <dbl>   <dbl>    <dbl>      <dbl>   <dbl>
1 ALASKA on time      497     221      212        503    1841
2 <NA>   delayed      62      12       20        102     305
3 <NA>   <NA>         NA      NA       NA         NA      NA
4 AM WEST on time     694    4840     383        320     201
5 <NA>   delayed     117     415      65        129      61
```

The data was recreated in a csv file and it is being pulled from a github repository.

```
df <- raw |> rename(airline = 1, status = 2)
df
```

```
# A tibble: 5 × 7
  airline status `Los Angeles` Phoenix `San Diego` `San Francisco` Seattle
  <chr>   <chr>      <dbl>   <dbl>    <dbl>      <dbl>   <dbl>
1 ALASKA on time      497     221      212        503    1841
```

2	<NA>	delayed	62	12	20	102	305
3	<NA>	<NA>	NA	NA	NA	NA	NA
4	AM WEST	on time	694	4840	383	320	201
5	<NA>	delayed	117	415	65	129	61

To make the data easier to work with, the first two unnamed columns are labeled.

```
clean <- df |>
  filter(!if_all(everything(), is.na)) |>
  fill(airline, .direction = "down") |>
  pivot_longer(
    cols = -c(airline, status),
    names_to = "city",
    values_to = "flights"
  ) |>
  pivot_wider(names_from = status, values_from = flights) |>
  rename(on_time = `on time`) |>
  mutate(
    total = on_time + delayed,
    delay_rate = delayed / total
  )

clean
```

```
# A tibble: 10 × 6
  airline city      on_time delayed total delay_rate
  <chr>   <chr>      <dbl>   <dbl> <dbl>      <dbl>
1 ALASKA Los Angeles    497     62   559      0.111
2 ALASKA Phoenix       221     12   233      0.0515
3 ALASKA San Diego     212     20   232      0.0862
4 ALASKA San Francisco  503    102   605      0.169
5 ALASKA Seattle     1841    305  2146      0.142
6 AM WEST Los Angeles   694    117   811      0.144
7 AM WEST Phoenix    4840    415  5255      0.0790
8 AM WEST San Diego    383     65   448      0.145
9 AM WEST San Francisco 320    129   449      0.287
10 AM WEST Seattle    201     61   262      0.233
```

The initial data table is large and has structural holes, hence, I clean it up in four main ways. Firstly, I removed the completely empty row that acts as a separator. Secondly, I populated the missing airlines by dragging the airline name downwards. Thirdly, I transformed the five city columns into long format. Lastly, I transformed the status to have total flights as a separate column for each airline and city.

```
overall <- clean |>
  group_by(airline) |>
  summarise(
    on_time = sum(on_time),
    delayed = sum(delayed),
    total = sum(total),
    delay_rate = delayed / total
  )

overall |>
  mutate(delay_rate = delay_rate)
```

```
# A tibble: 2 × 5
  airline on_time delayed total delay_rate
<chr>    <dbl>    <dbl> <dbl>    <dbl>
1 ALASKA   3274      501  3775    0.133
2 AM WEST  6438      787  7225    0.109
```

Alaska is delayed about 13.3% of the time and AM West about 10.9%, so according to the provided data, AM West looks more reliable.

```
by_city <- clean |>
  select(airline, city, delay_rate) |>
  pivot_wider(names_from = airline, values_from = delay_rate)
by_city |>
  mutate(across(c(ALASKA, `AM WEST`)))
```

```
# A tibble: 5 × 3
  city          ALASKA `AM WEST`
<chr>         <dbl>    <dbl>
1 Los Angeles  0.111     0.144
2 Phoenix     0.0515    0.0790
3 San Diego   0.0862    0.145
4 San Francisco 0.169    0.287
5 Seattle     0.142    0.233
```

Alaska has the lower delay rate in all five cities, which goes against the previous conclusion we drew from when we looked at the overall delay rates of just the airlines.

```
clean |>
  group_by(airline) |>
  mutate(share_of_flights = total / sum(total)) |>
```

```
ungroup() |>
select(airline, city, share_of_flights) |>
pivot_wider(names_from = airline, values_from = share_of_flights)
```

```
# A tibble: 5 × 3
  city      ALASKA `AM WEST`
  <chr>      <dbl>      <dbl>
1 Los Angeles 0.148      0.112
2 Phoenix    0.0617     0.727
3 San Diego  0.0615     0.0620
4 San Francisco 0.160     0.0621
5 Seattle    0.568     0.0363
```

The majority of the flights that Alaska flies are towards Seattle, which has relatively high delays for both the airlines whereas, for AM West, the majority of its flights are directed to Phoenix, which rarely sees any delay for either of the airlines. AM West's overall average has been dragged down by its Phoenix flights despite being inferior to Alaska at every single airport.

Conclusion

AM West has the smaller overall percentage of delays, but Alaska has the smaller delay percentage at each of the five airports. From the point of view of an individual who will choose an airline to fly in a certain route, the comparison at the city level is the correct one, favoring Alaska, since the overall statistic mainly represents where each airline operates. The sample considered is only a snapshot of one moment in time and includes only five airports. Furthermore, there is no explanation for what makes delays different between airlines as weather and hub issues could be causes but were not checked here.